VAMOS: A Framework for Fast, Configurable and Accessible Multiobjective Optimization in Python

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Abstract

Python is widely used for empirical research in multi-objective optimization due to its accessibility and strong scientific ecosystem. However, most Python frameworks store populations as per-solution objects and keep critical computations in the interpreter, which prevents vectorization and significantly limits performance. This design gap makes Python-based metaheuristics much slower than their counterparts in compiled languages. In this paper we introduce VAMOS (Vectorized Architecture for Multiobjective Optimization Suite), a vectorized Python framework for multi-objective evolutionary algorithms (MOEAs). VAMOS maintains all population data in dense numerical arrays and dispatches hot-path operations to interchangeable numerical backends, decoupling algorithm logic from the underlying compute substrate. The framework implements eight MOEAs spanning dominance-based, decomposition-based, and indicator-based paradigms, and exposes rich component-level configurability, including interchangeable operators, external archives, and fine-grained parameter settings. VAMOS also integrates an automated algorithm-configuration module and provides a browser-based graphical interface to simplify use by non-experts. The results show that VAMOS achieves statistically equivalent levels of solution quality than those elicited by MOEAs implemented in state-of-the-art libraries, while reducing wall-clock time by large margins across all tested algorithms. These gains confirm that vectorized population models and backend-agnostic kernels at the core of VAMOS can deliver high performance in Python without compromising algorithmic behavior.

Keywords: Evolutionary algorithms, Multiobjective optimization, Software framework, Vectorization

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